Tabulation

Ques: Longest Increasing Subsequence (LIS)

LIS = 6

[0, 8, 4, 12, 2, 10, 6, 14, 1, 9, 5, 13, 3, 11, 7, 15]

dp	1	2	2	3	2	3	3	4	2	4	3	5	3	5	4	6

dp[i] will store the length of LIS upto that pt. including that number

arr = 7 2 6 5 8 10

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Without DP. 'Binary Search'

Ques: Longest Increasing Subsequence (LIS)

LIS = 6

[0, 8, 4, 12, 2, [redacted] 6, [redacted] 1, 9, 5, 13, 3, 11, 7, 15]

ans = 0 8 12 7 9 13 15

4 9 7 11

2 8

1 3

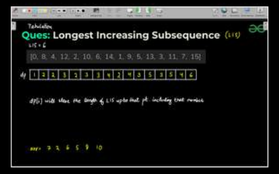
Only Length

This ans array will always be sorted.

lower bound of an ele x is arr[i] ≥ x

↓

replace



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Reasons to Replace ele by its lower bound.

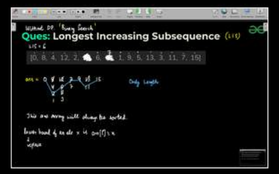
1) 1 2 7 3 4

1 2 ~~7~~ 4
3

2) 0 8 4 7 2 3 5

0 8 7
4
2

2 is not disturbing 7's LIS (the LIS which ends at 7)
rather it is replaced for a maybe better solution



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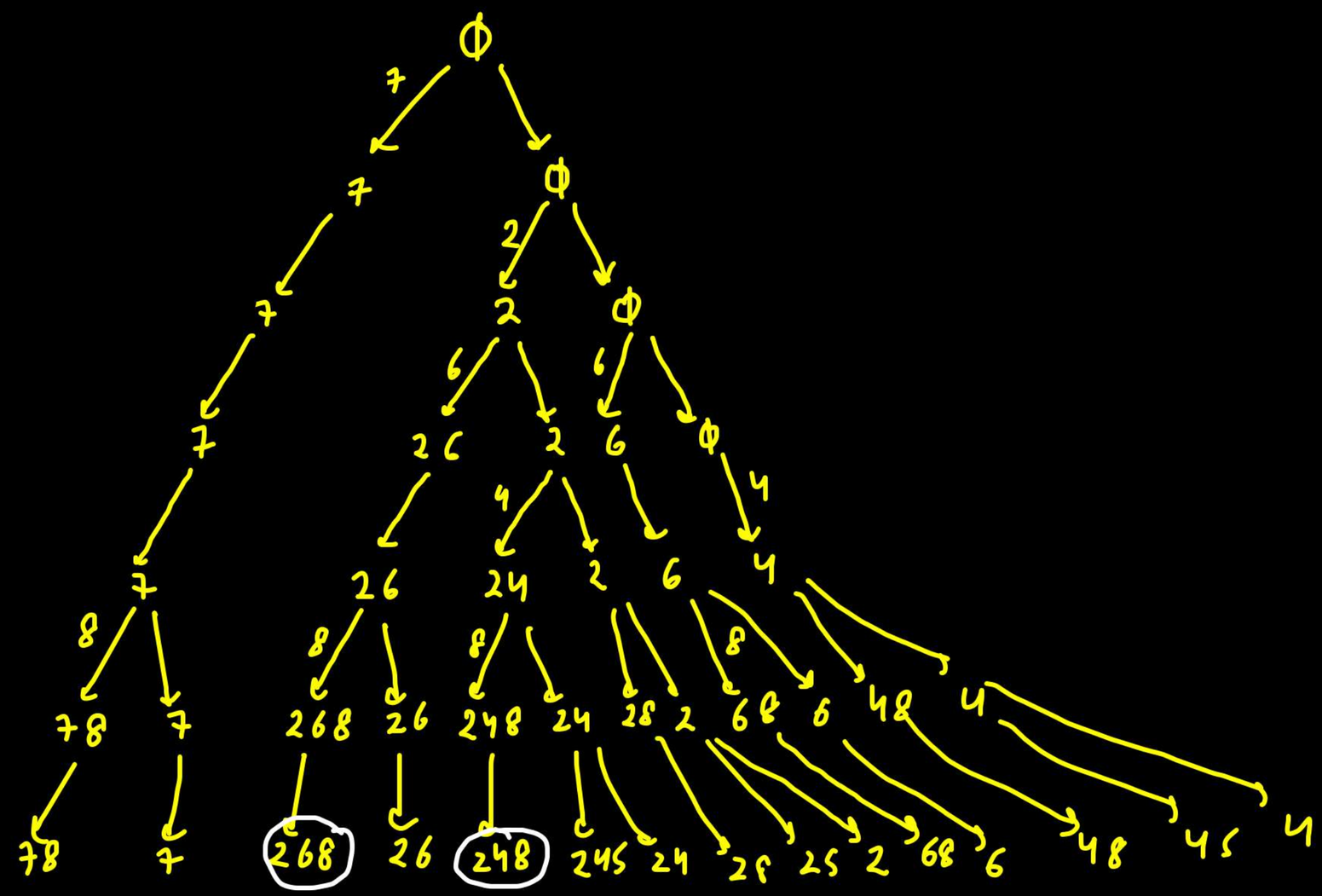
arr = 7 2 6 4 8 5

pick → length ++

skip

max(pick, skip)

fun(idx, prevIdx)



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Ques: Longest Bitonic Subsequence

1 11 2 10 4 5 2 1

LIS & LDS Combined

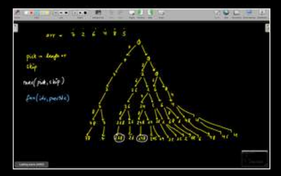
✓ 1 2 10 5 2 1

len = 6

✓ 1 2 4 5 2 1

✓ 1 11 10 4 2 1

Strictly ↑ & ↓ are not considered bitonic



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arr	1	11	2	10	4	5	2	1

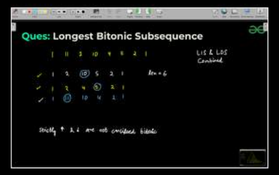
dp1	1	2	2	3	3	4	2	1

LIS from left to right

dp2	1	5	2	4	3	3	2	1

1 2 10

10 5 2 1



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Ques: Largest Divisible Subset (Hint LIS)

arr = 1 27 21 81 3 7 100

{1 27 81 3} → 4

{1 3 21} → 3

{1 7 21} → 2

Sort → 1 3 7 21 27 81 100

dp 1 2 2 3 3 4 2

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arr = 1 2 4 8 6 30

1 2 4 8 ✗

1 2 6 30 ✓

Think about LDS

reverse loop

arr = 2 14 126 3 15 105 ⟶

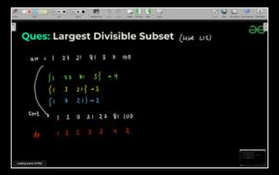
	2	3	14	15	105	126	
			✗	✓	✓		
dp1	1	1	2	2	3	3	✗
dp2	3	3	2	2	1	1	✓

2 14 126

3 15 105 ✓

maxLen = 3

{3, 15, 105}



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arr = 2 3 14 15 51 105 126 561

dp2 = 3 3 2 2 2 1 1 1

Sort this

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Ques: Russian Doll Envelopes

(2, 3), (5, 9), (6, 1), (6, 7), (7, 2), (8, 3), (9, 4)

3 9 1 7 2 3 4

(2, 3), (5, 9), (6, 7), (6, 2), (6, 1), (8, 3), (9, 5) (9, 4)

3 9 7 2 1 3 5 4

can never be part of a LIS together

LIS = 3